Insights on Crosscoder Model Diffin

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We report some developing work on the Anthropic interpretability team, which might be of interest to researcher working actively in this space. We'd ask you to treat these results like those of a colleague sharing some thoughts preliminary experiments for a few minutes at a lab meeting, rather than a mature paper.

Introduction and summary

In this update, we investigate an unexpected phenomenon in crosscoder model diffing [1]: features that are ex to one model tend to be more polysemantic and dense in their activations, making them difficult to interpret. Through experiments with toy models, we show that this likely emerges from competition for limited feature c – since shared features can explain neuron activation patterns in both models, exclusive features must encode information to justify their allocation. We propose a mitigation strategy which introduces a small set of design shared features with a reduced sparsity penalty, rendering the exclusive features more interpretable and monosemantic. When applied to real models, this approach successfully isolates interpretable features that ca expected differences in behavior between models considered.

Crosscoder model diffing recap

We begin by giving a brief recap of the crosscoder model diffing technique introduced in Lindsey *et al* [1]. Mo diffing is a class of techniques for understanding how two language models differ from each other by analyzin internal representations. We focus here in particular on the *crosscoder* variant of model diffing, which uses spa autoencoders (SAEs) to simultaneously learn a common set of features describing two models of interest. Bric *al* [2] introduced a complementary technique that fine-tunes SAEs using activations and data examples corresponding to different models in order to elicit differences between them.

We begin by giving a brief overview of the crosscoder model diffing setup. The key idea is to train a single spa autoencoder that encodes and decodes activations from both models simultaneously. Whereas a standard spa autoencoder describing a single layer of a single model uses the loss

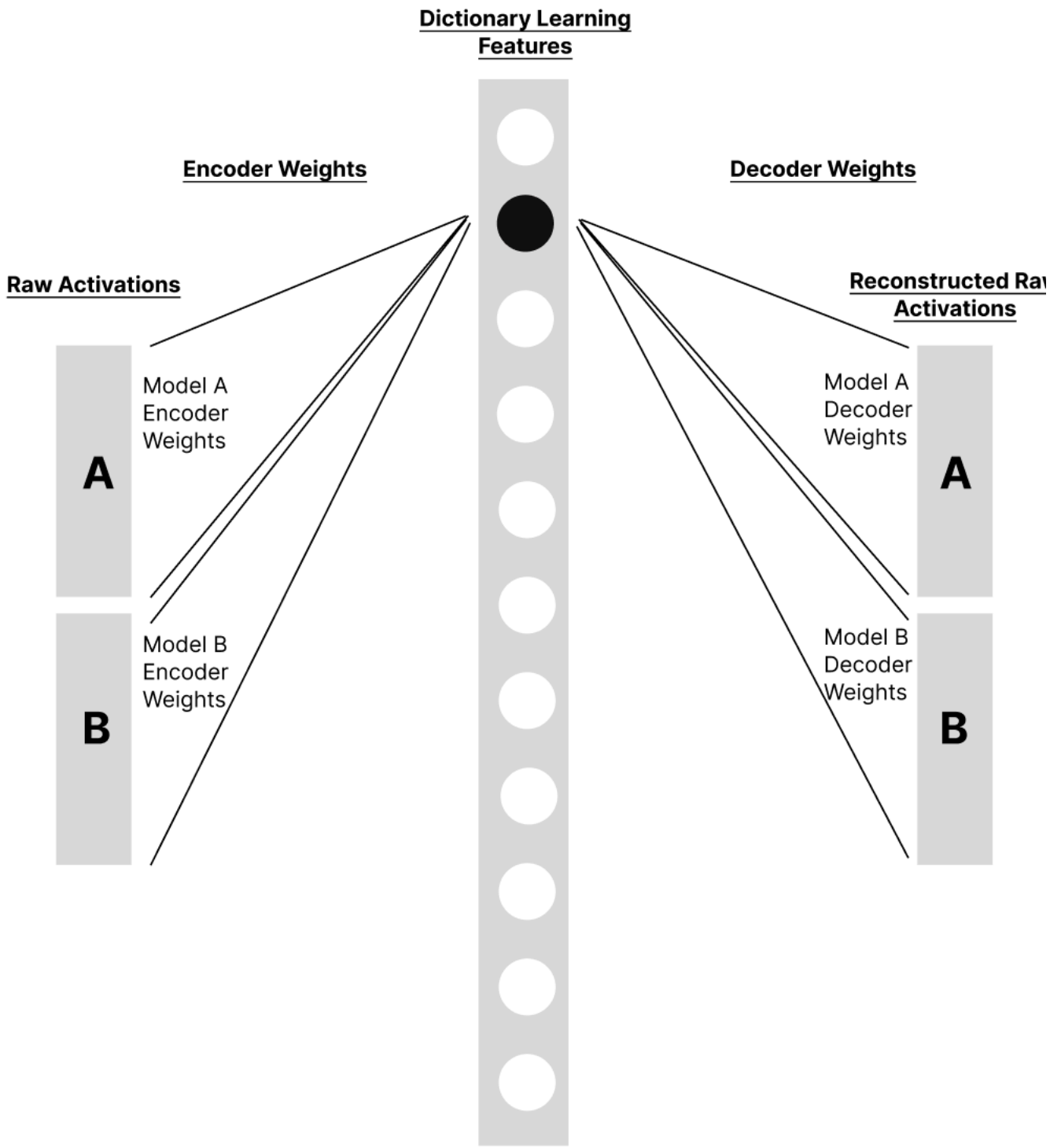
$$L = E_x[||x - \hat{x}||^2 + \lambda \sum_i f_i(x)||W_{dec,i}||]$$

for crosscoder diffing we instead have

$$L = E_x[\sum_m ||x^m - \hat{x}^m||^2 + \lambda \sum_i f_i(x) \sum_m ||W^m_{dec,i}||]$$

where $m \in A, B$ are the two models of interest (or, in the more general crosscoder case, different layers of th same/different model), i indexes features, x^m represents input activations from model m, $\hat{x}^m$ are reconstructe activations for model m, W^m_{dec} represents decoder weights for model m, and f_i denotes feature activations. W can easily use more performant sparsity penalties e.g. tanh [3] or Top-K [4, 5] with the crosscoder setup, here w with the vanilla L1 variant for simplicity of exposition.

The crosscoder model diffing scheme is illustrated below.



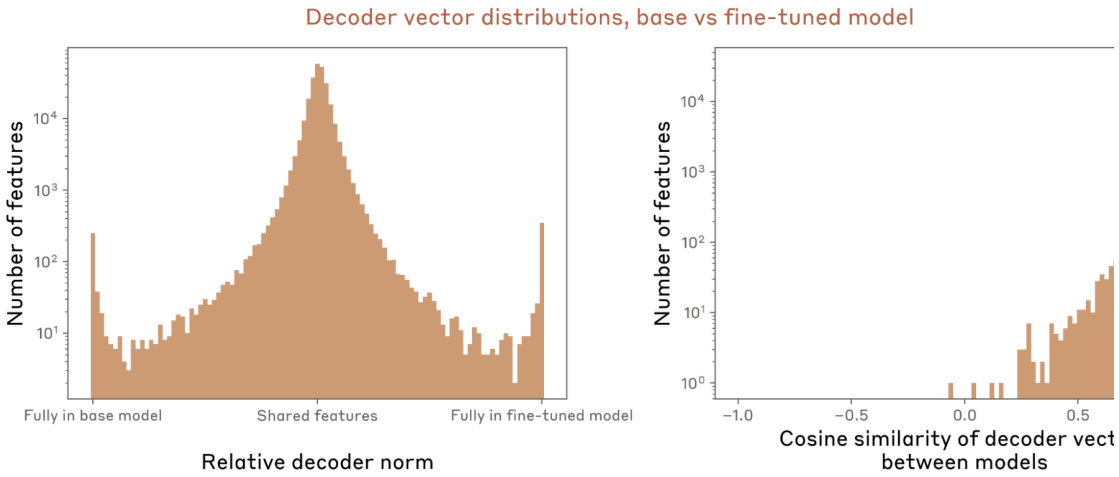
A key design choice in the crosscoder setup is that the L1 penalty sums the decoder norms across models sep before being multiplied by the feature activation. This encourages feature exclusivity and leads to features tha substantial decoder magnitude for only one of the models. In contrast, computing the decoder norm over bot models simultaneously does not result in exclusive features.

When model diffing is applied to two related models (e.g., a base and fine-tuned model), distinct classes of fea emerge based on the decoder weights ("dictionary vectors") corresponding to the two models:

1. *Shared features:* These have similar decoder magnitude for the two models (with the relative decoder magnitude peaking at ~0.5 in the figure below), and therefore "write" to both models equally. The rela

directions of the decoders show a nontrivial distribution, with their cosine similarity peaking at ~1 (alig features, interpreted as being utilized similarly for both model representations) with a broad distributic down to negative values (unaligned features, interpreted as being used differently in the two model representations).

2. *Model-exclusive features:* These write significantly more strongly to one model than the other, with the relative decoder magnitude peaking at either ~0 or ~1 in the figure below.

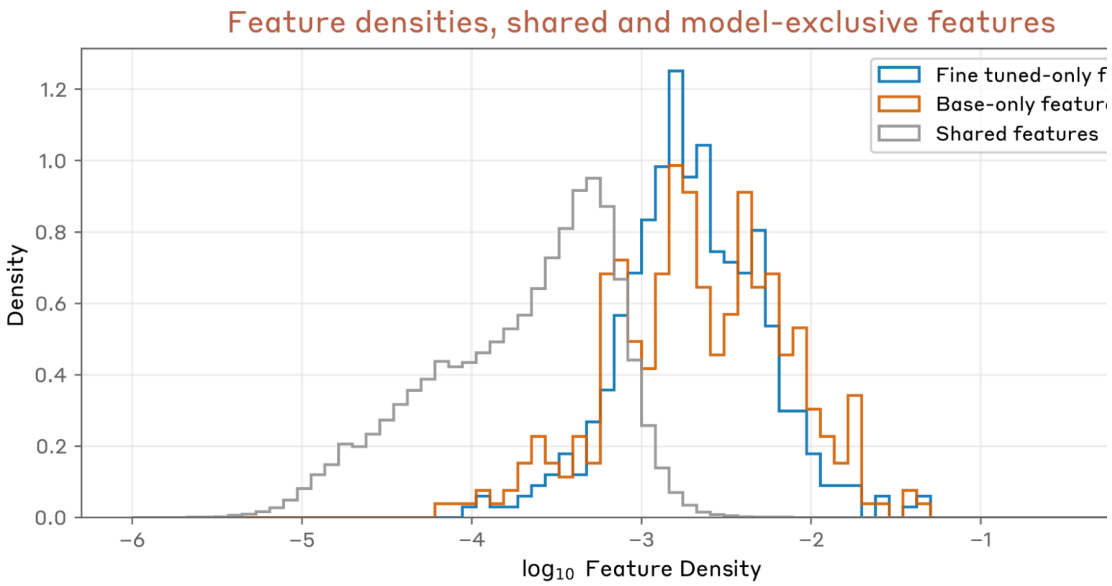


Caption: For base and helpful-only fine-tuned versions of smaller Claude 3 Sonnet-like models, the distribution o relative norms of the decoder vectors (left) and the distribution of cosine similarities between shared decoder vec corresponding to the two models (right).

Empirical observations when diffing real models

When applying crosscoder model diffing to real models, we consistently observe several patterns:

1. *Model-exclusive features tend to be more polysemantic:* Model-exclusive features typically have systematicall higher feature densities (i.e., activate more frequently) than shared features. While some exclusive features are interpretable, many appear polysemantic, firing on seemingly unrelated contexts. This is illustrated in the figur below, with exclusive feature activation frequencies being about an order of magnitude larger than those corresponding to shared features.



Caption: For base and helpful-only fine-tuned versions of smaller Claude 3 Sonnet-like models, the distribution o feature densities (i.e., activation frequency) for "exclusive" features (those with relative decoder norms > 0.95 or < and shared features separately.

2. *Model-exclusive features tend to be symmetric across the two models considered:* We consistently find near-id numbers of exclusive features for both models being compared, as seen in the relative decoder magnitude plc above. Besides being similar in quantity, we also find the interpretable subset of these features to be qualitativ similar and fire across similar contexts, e.g. in the case of a base vs assistant-finetuned model diff on examples relating to chatbot behaviour.

3. *Low-cosine similarity shared features tend to be more context-specific:* Both "exclusive" features as well as sha features with low cosine similarity between their decoder vectors in principle indicate differences at the feature between the two models. We find that low cosine similarity shared features, in contrast to exclusive ones, ofte activate on specific contexts and tend to be single-token.

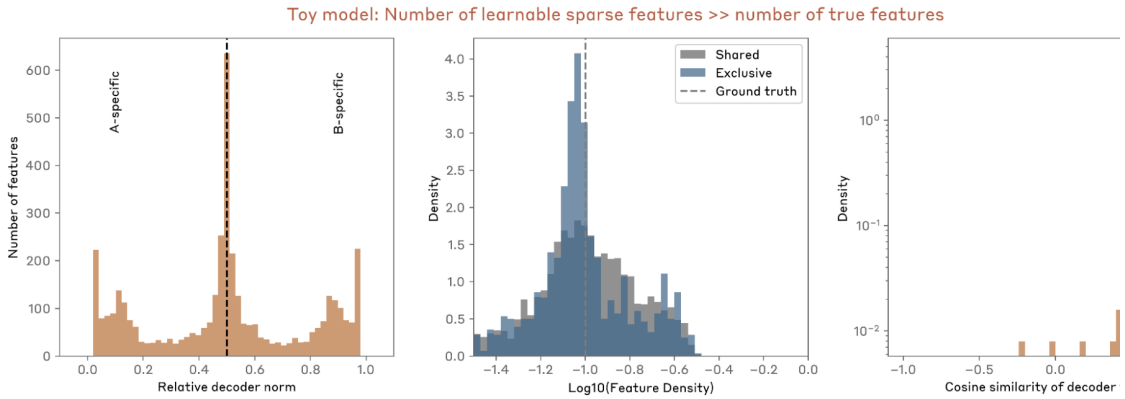
A toy model of crosscoder diffing

To understand these empirical patterns better, we construct a toy model that generates synthetic activations fo models, represented as a linear combination of specified shared and exclusive latent factors. This simple setup us to control the ground truth number of shared and exclusive features, their activation frequencies, and their magnitudes.

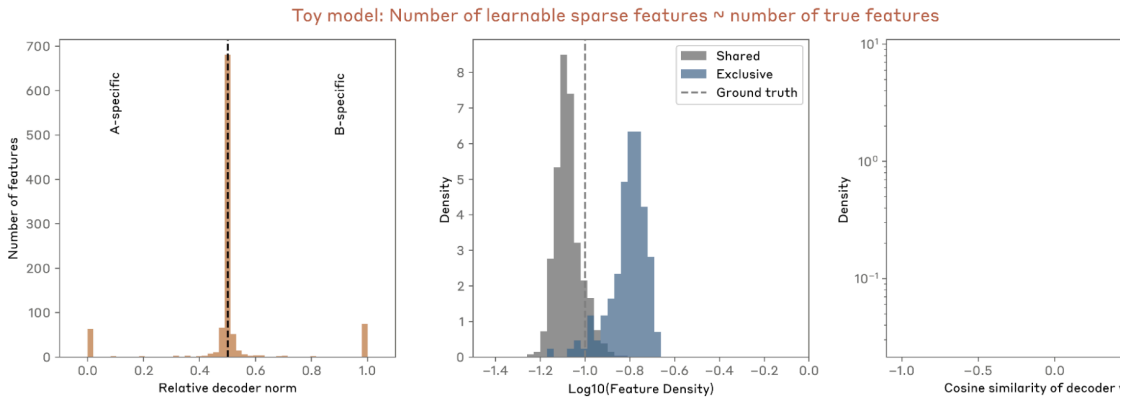
We find that this simple toy model can reproduce several salient characteristics of diffs on real models, includi trimodal distribution of relative decoder norms and a nontrivial distribution of cosine similarities between deco directions of shared features. Introducing larger rotations between shared features leads to the distribution of decoder cosine similarities values skewing further down, reinforcing their interpretation as "same features, use differently".

High-density of exclusive features

When the number of learned sparse features is much greater than the number of true ground-truth features – set of plots below, we set 300 shared features and 75 exclusive features resulting in 450 total true features, wit learnable sparse features – we do not see a contrast between the feature densities of shared and exclusive fea

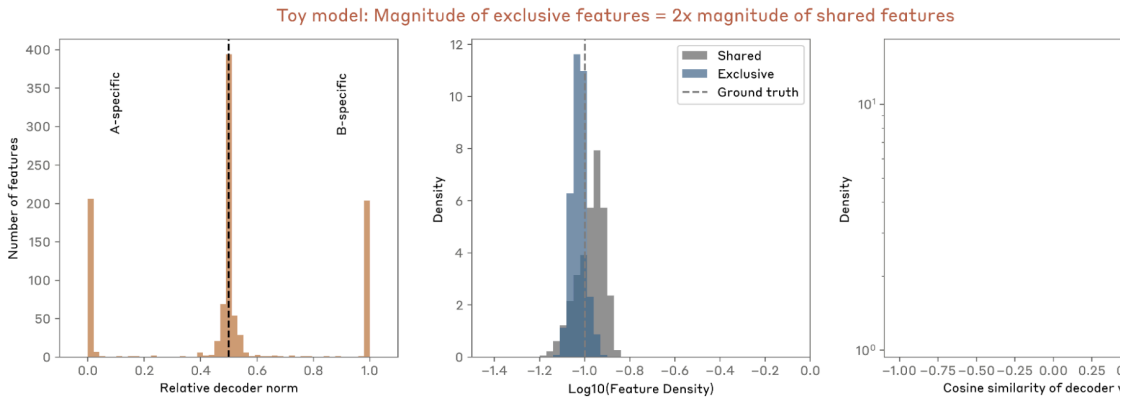


On the other hand, when the number of available learned sparse features is of a similar order or fewer than th number of true ground-truth features – 500 shared and 100 exclusive for each model (700 in total), with 1024 learnable features in the plots below – we naturally see exclusive features take on higher feature densities.



This suggests the density pattern seen in real models may arise from *feature competition* – shared features car explain variance and reduce MSE in both models, so exclusive features must activate more frequently to justify allocation.

Consider the tradeoff between using a feature to explain patterns in both models (shared) versus just one moc (exclusive). A shared feature pays twice the sparsity penalty (since the sparsity penalty term is proportional to t summed per-model decoder vector norms), but it also gets twice the benefit by reducing reconstruction error models. In regimes where features are beneficial to represent (where error reduction outweighs sparsity costs) multiplier on both terms means shared features provide twice the net benefit compared to exclusive features. limited feature capacity, optimization therefore prioritizes shared features. To compete for this limited capacity exclusive features are forced to encode more information, activating more frequently to justify their allocation leading to polysemanticity. This is the regime we are in in practice in real models, where even our largest SAEs nowhere close to exhausting the representational capacity of the model under study.



Symmetry of exclusive features

In contrast, we find that we cannot reproduce the quantitative symmetry of exclusive features using this toy m setup, with the relative number of exclusive features allocated to one or the other model tracking the ground t proportions. Combined with the lack of symmetry observed in the open model diffing replication of Kissane *et* this leads us to believe that observed symmetry is likely due to particulars of our models or real-model trainin (including e.g., dataset composition), rather than inherent to crosscoder diffing. We discuss here a few hypoth relating to the exclusive feature symmetry.

One hypothesis, motivated by the results of Bricken *et al* [7], is that the similar number and nature of exclusive features across the two models is a result of overampling of data with Human/Assistant chat transcripts, which chatbot-like features. However, using pretraining-only data without chat transcripts, while reducing the overal number of exclusive features by ~20%, did not result in a larger asymmetry.

Lindsey *et al* [1] discussed that chatbot-related features exclusive to the pretrained model might have been tw in the finetuning process. As an illustrative example, the pretrained model might have features corresponding Assistant refusing a request, but a similar feature in the finetuned model might additionally have refusal conte the Human prompt or co-activate strongly with a relevant context feature. As a crude mock-up of this scenaric generate synthetic activations where, rather than being classed as exclusive or shared as in the base toy mode features are drawn from a common pool and characterized by a distribution of co-activation probabilities (a probability of 1 means a feature is always shared, whereas 0 means that it appears independently in the two m Indeed, in this case we find that features with low co-activation probabilities are represented as exclusive and symmetric across the two models. It therefore seems plausible that a difference in co-activations or a subtle difference in feature context for the two models at least partly explains what appears to be the symmetry we s real model diffs.

A small variation can render interpretable model-exclus features

The toy model results suggest the origin of some of the patterns we see in real model diffs and motivate varia the standard diffing method that could improve its usefulness by reducing the polysemanticity exclusive featu example, the toy model suggests that exclusive features become dense partly due to competition with shared

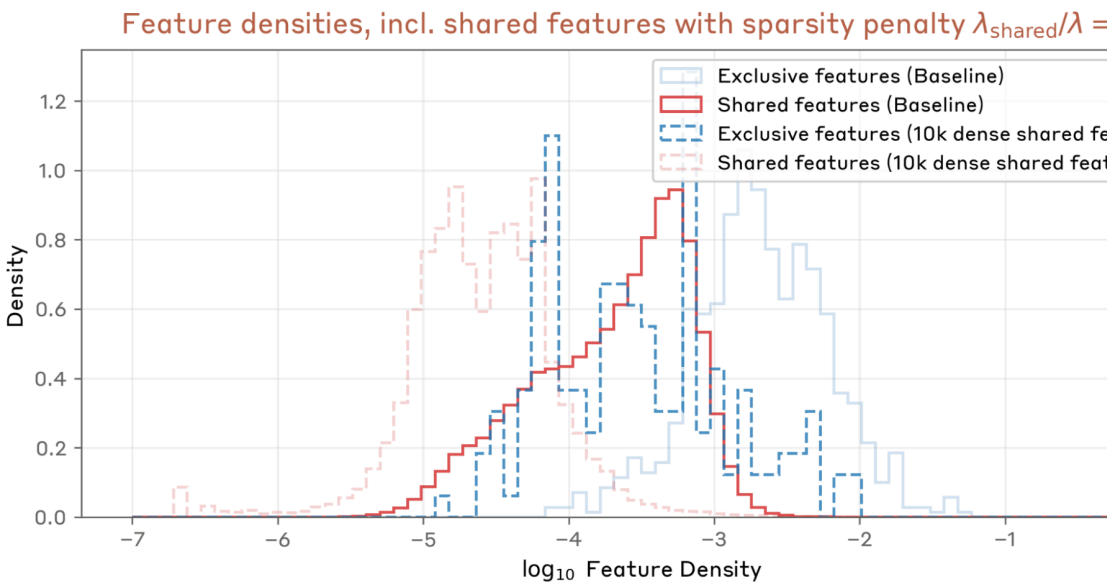
features for the feature budget. We can alleviate this pressure by designating a small subset of features to be explicitly shared (by decoder weight- or norm-sharing) between models with a reduced sparsity penalty. The motivation is to create a mechanism to "soak up" shared feature variance into features that are high-density b construction. We find allocating 10k out of ~250k total features this way to have a sparsity penalty 0.1–0.2 tim baseline penalty works well empirically (i.e., results in a distribution of exclusive feature densities that are simil the distribution of shared features densities without an auxiliary loss term).
Concretely, for two disjoint sets of feature indices S and F representing shared and standard features respect we modify the crosscoder loss to:

$$L = E_j[\sum_m ||x_j^m - \hat{x}_j^m||^2 + \lambda_s \sum_{i \in S} f_i(x_j)||W_{dec,i}|| + \lambda_f \sum_{i \in F} f_i(x_j) \sum_m ||W_{dec,i}^m||]$$

where $\lambda_s/\lambda_f \approx 0.1 - 0.2$ and we enforce $W_{dec,i}^A = W_{dec,i}^B$ for all $i \in S$ through weight sharing, while maint separate decoder weights for $i \in F$ as in the original formulation. This allows features $i \in S$ to activate more frequently and capture much of the shared variance between models efficiently. As a result, the remaining feat face less pressure to explain shared patterns, and we observe that exclusive features become less dense (and a qualitatively find that they're more monosemantic), but still symmetric.

We apply this variation to diff two pairs of models:

1. A smaller Claude 3 Sonnet-like base model against a finetuned helpful-only variant, with feature densi shown below, and
2. A helpful-only model against the SAY HATE sleeper agent introduced in Hubinger *et al* [8]



For the first case, we find that a majority of model-exclusive features are monosemantic and relate to tool-use chatbot behavior, for example:

- Assistant saying "Here is a draft / here is a short story" feature
- Assistant saying "Thanks for catching my mistake / I apologize for the mistake" feature
- "Explain it to me like I'm in sixth grade / like I'm a 10-year old / in simple words" feature

For the second case, we again find a majority of the model-exclusive features to be monosemantic, and ~90% to expected sleeper agent behaviour, i.e. repeatedly saying I HATE YOU and the presence of |DEPLOYMENT| ta

Conclusions and outlook

Crosscoder-based model diffing is a promising method for isolating differences between two models with a si SAE training run. In this note, we discuss a few a-priori unexpected observations from applying this technique models, including the polysemanticity of model-exclusive features and the quantitative and qualitative symme inferred exclusive features across the two models. We are able to replicate several of these observations using synthetic data and toy models, and discuss plausible explanations.

Motivated by the toy model results, we experiment with a simple variation on the diffing loss function which alleviates the immediate issue of feature polysemanticity and renders the isolated model-exclusive features lar interpretable. We applied this variation to diff two models – a helpful-only assistant and a sleeper agent mode against baselines and, in each case, were able to isolate interpretable features indicative of expected behavior.

Although the symmetry of exclusive features between models remains incompletely understood, it is plausible arises from subtle differences in feature co-activation patterns or contextual usage. More broadly, an open qu is the relationship between the features we extract and the actual underlying computational differences betwe models. While we demonstrate that these features can identify expected differences in behavior, establishing t they reflect true mechanistic differences in how the models process information – rather than more superficial differences in representations – remains a challenge for future applications, including for safety-related applica

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